

Multi-Agent RAG Workflow: Execution Trace

This document provides a step-by-step execution trace of a multi-part user query flowing through the Multi-User Multi-Domain Agentic RAG System. The query requires accessing both HR and IT domains, demonstrating the interaction between the Supervisor, Query-Planning, Retrieval, Verification, and Synthesis agents.

Step 0: Initialization (Query Entry)

- **Event:** The system initializes the starting state and passes it to the Supervisor. The Auth system has verified that the user has permission to access the HR and IT domains.

```
{
  "messages": [
    { "role": "user", "content": "Based on the HR policy documents,
how many remote work days are allowed, and what is the current
status of my hardware request in the IT ticketing system?" }
  ],
  "next_agent": null,
  "step_count": 0,
  "allowed_domains": ["HR", "IT"],
  "target_domain": null,
  "plan": null,
  "retrieved_context": [],
  "is_context_valid": null,
  "verification_feedback": null,
  "citations": null
}
```

Step 1: Supervisor Decides to Plan

- **Event:** The Supervisor reads the query, determines it contains multiple parts, and routes the state to the Query-Planning Agent.
- **State Updates:** Increment `step_count` and update `next_agent`.

```
{
  ... // existing state
```

```
"step_count": 1,  
"next_agent": "Query-Planning_Agent"  
}
```

Step 2: Query-Planning Agent Creates the Plan

- **Event:** The Query-Planning Agent receives the State, analyzes the query, writes a step-by-step plan, and hands control back to the Supervisor.
- **State Updates:** Populate the plan array and reset next_agent to Supervisor.

```
{  
  ... // existing state  
  "plan": [  
    "Task 1: Search HR policies for allowed remote work days.",  
    "Task 2: Search IT system records for hardware request status."  
  ],  
  "next_agent": "Supervisor",  
  "step_count": 1  
}
```

Step 3: Supervisor Delegates First Task (HR Retrieval)

- **Event:** The Supervisor reads the first task, routes it to the Retrieval Agent, and restricts the search scope strictly to Human Resources.
- **State Updates:** Set target_domain to "HR", update next_agent, and increment step_count.

```
{  
  ... // existing state  
  "target_domain": "HR",  
  "next_agent": "Retrieval_Agent",  
  "step_count": 2  
}
```

Step 4: Retrieval Agent Fetches HR Data

- **Event:** The Retrieval Agent searches Qdrant using the metadata filter (domain = HR). It finds the relevant policy, appends it to the context, and returns control to the Supervisor.
- **State Updates:** Append the fetched chunk to retrieved_context.

```
{
  ... // existing state
  "retrieved_context": [
    {
      "chunk_id": "chk_hr_992",
      "file_name": "HR_Remote_Policy.pdf",
      "text": "Full-time employees are allowed to work remotely for
a maximum of two days per week.",
      "page": 2
    }
  ],
  "next_agent": "Supervisor",
  "step_count": 2
}
```

Step 5: Supervisor Requests Context Verification

- **Event:** The Supervisor routes the newly retrieved context to the Verification Agent to assess its quality and relevance.
- **State Updates:** Update next_agent and increment step_count.

```
{
  ... // existing state
  "next_agent": "Verification_Agent",
  "step_count": 3
}
```

Step 6: Verification Agent Approves the Result

- **Event:** The Verification Agent determines the context adequately answers the first task. It approves the context and routes back to the Supervisor.
- **State Updates:** Set is_context_valid to true.

```
{
```

```

... // existing state
"is_context_valid": true,
"verification_feedback": null,
"next_agent": "Supervisor",
"step_count": 3
}

```

Step 7: Supervisor Delegates Second Task (IT Retrieval)

- **Event:** Seeing task 1 is resolved, the Supervisor moves to the second task in the plan. It routes to the Retrieval Agent again, but this time shifts the domain to Information Technology.
- **State Updates:** Set target_domain to "IT", reset is_context_valid to null, update next_agent, and increment step_count.

```

{
... // existing state
"target_domain": "IT",
"is_context_valid": null,
"next_agent": "Retrieval_Agent",
"step_count": 4
}

```

Step 8: Retrieval Agent Fetches IT Ticket Status

- **Event:** The Retrieval Agent searches Qdrant using the metadata filter (domain = IT) across text and vector records. It appends the ticket status to the shared context and returns control.
- **State Updates:** Append the new chunk to retrieved_context.

```

{
... // existing state
"retrieved_context": [
{
"chunk_id": "chk_hr_992",

```

```

      "file_name": "HR_Remote_Policy.pdf",
      "text": "Full-time employees are allowed to work remotely for
a maximum of two days per week.",
      "page": 2
    },
    {
      "chunk_id": "chk_it_req_5541",
      "file_name": "IT_Tickets_Dump.json",
      "text": "Hardware Request #5541: Approved and currently
shipping.",
      "page": null
    }
  ],
  "next_agent": "Supervisor",
  "step_count": 4
}

```

Step 9: Supervisor Directs to Final Synthesis

- **Event:** The Supervisor verifies that all tasks in the plan are complete and the required data is gathered. It directs the workflow to the Synthesis Agent.
- **State Updates:** Update next_agent and increment step_count.

```

{
  ... // existing state
  "next_agent": "Synthesis_Agent",
  "step_count": 5
}

```

Step 10: Synthesis Agent Drafts Final Answer (Final Output)

- **Event:** The Synthesis Agent formulates the final answer in English, maps the specific statements to their source citations, and ends the workflow graph.
- **State Updates:** The assistant's response is appended to messages, citations are mapped, and next_agent is set to "END".

```
{
  "messages": [
    {
      "role": "user",
      "content": "Based on the HR policy documents, how many remote work days are allowed, and what is the current status of my hardware request in the IT ticketing system?"
    },
    {
      "role": "assistant",
      "content": "Based on company policies, you are allowed a maximum of two remote work days per week. Regarding your hardware request, it has been approved and is currently shipping."
    }
  ],
  "next_agent": "END",
  "step_count": 5,
  "allowed_domains": ["HR", "IT"],
  "target_domain": "IT",
  "plan": [
    "Task 1: Search HR policies for allowed remote work days.",
    "Task 2: Search IT system records for hardware request status."
  ],
  "retrieved_context": [
    {
      "chunk_id": "chk_hr_992",
      "file_name": "HR_Remote_Policy.pdf",
      "text": "Full-time employees are allowed to work remotely for a maximum of two days per week.",
      "page": 2
    },
    {
      "chunk_id": "chk_it_req_5541",
      "file_name": "IT_Tickets_Dump.json",
      "text": "Hardware Request #5541: Approved and currently shipping.",
      "page": null
    }
  ],
  "is_context_valid": null,
  "verification_feedback": null,
  "citations": [
    {
```

```
    "span": "allowed a maximum of two remote work days per week",
    "source_file": "HR_Remote_Policy.pdf",
    "chunk_id": "chk_hr_992",
    "page": 2
  },
  {
    "span": "has been approved and is currently shipping",
    "source_file": "IT_Tickets_Dump.json",
    "chunk_id": "chk_it_req_5541",
    "page": null
  }
]
```